A POINT IS CLOSED IN ITS FIBRE EXACTLY WHEN ITS RESIDUE FIELD EXTENSION OVER THE BASE IS FINITE

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 3596-3604 and proof span 3606-3630 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof's closing phrase 'we win' is the author's and is preserved verbatim.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A point of a fibre whose residue field extension over the base is finite is a closed point of that fibre, cited from the same chapter. Cited at proof lines 3823-3825. Gives one direction of the equivalence directly.
- The same-chapter characterization of locally finite type morphisms on affine charts, so that the corresponding ring map is of finite type. Cited at proof lines 3829-3830. Makes the fibre ring a finite type algebra over the residue field of the image point, which is what the Nullstellensatz needs.
- Hilbert's Nullstellensatz in the form that a maximal ideal of a finite type algebra over a field has finite residue field extension, cited from the algebra chapter. Cited at proof lines 3840-3842. Converts the maximality of the prime of the fibre ring into finiteness of the residue field extension.

Source reference. Original theorem 00FV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 585–585. Same-chapter reference used in the retained author text at source line(s) 3624. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01TE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2371–2371. Same-chapter reference used in the retained author text at source line(s) 3609, 4034, 5665, 8205. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Same-chapter reference used in the retained author text at source line(s) 2717, 2739, 2801, 2863, 3019, 3035, 3157, 3348, 3613, 3707, 3868, 5197, 5282, 6702, 9325, 9390, 9440, 11421, 13422, 13686. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ be a point. Set $s = f(x)$. Assume f is locally of finite type. Then x is a closed point of its fibre if and only if $\kappa(x)/\kappa(s)$ is a finite field extension.*

Proof. If the extension is finite, then x is a closed point of the fibre by Lemma 01TE above. For the converse, assume that x is a closed point of its fibre. Choose affine opens $\mathrm{Spec}(A) = U \subset X$ and $\mathrm{Spec}(R) = V \subset S$ such that $f(U) \subset V$. By Lemma 01T2 the ring map $R \to A$ is of finite type. Let $\mathfrak{q} \subset A$, resp. $\mathfrak{p} \subset R$ be the prime ideal corresponding to x, resp. s. Consider the fibre ring $\overline{A} = A \otimes_R \kappa(\mathfrak{p})$. Let $\overline{\mathfrak{q}}$ be the prime of $\overline{A}$ corresponding to $\mathfrak{q}$. The assumption that x is a closed point of its fibre implies that $\overline{\mathfrak{q}}$ is a maximal ideal of $\overline{A}$. Since $\overline{A}$ is an algebra of finite type over the field $\kappa(\mathfrak{p})$ we see by the Hilbert Nullstellensatz, see Algebra, Theorem 00FV, that $\kappa(\overline{\mathfrak{q}})$ is a finite extension of $\kappa(\mathfrak{p})$. Since $\kappa(s) = \kappa(\mathfrak{p})$ and $\kappa(x) = \kappa(\mathfrak{q}) = \kappa(\overline{\mathfrak{q}})$ we win. □